

DASE7501 Robot Modelling, Planning and Control —Focus Paper

(Lectures 1, 2 & 4)

Modelled on Homework 1 and past-exam styles • 20 questions • 100 marks • Suggested time 60 minutes

Part 1: Questions

Instructions: This focus paper covers the examinable points of Lectures 1, 2 and 4, and is modelled on Homework 1 and the question styles tested in past exams (applications, DOF, robot selection, accuracy/repeatability, sensitivity, PWM). It is written in English, exactly like the real exam. Answer ALL questions first; only when you finish should you open Part 2 (Answer Key and Solutions), which is at the end of this document. No Chinese annotations appear in this part.

Time suggestion: 60 minutes. **Total:** 100 marks.

Section A: Multiple Choice (10 questions, 2 marks each, 20 marks)

Q1. Which of the following is NOT a typical application of industrial robots in manufacturing and logistics?

- A. Welding
- B. Painting
- C. Assembly
- D. Cooking meals in a restaurant kitchen

Q2. To polish a free-form surface, the minimum number of degrees of freedom (DOF) that a robot must have is:

- A. 3
- B. 4
- C. 5
- D. 6

Q3. A SCARA robot has three revolute joints with parallel vertical axes and one prismatic joint that moves vertically. Its number of DOF is:

- A. 3
- B. 4
- C. 5
- D. 6

Q4. Which robot type is best suited for handling very large parts and moving material over long distances?

- A. SCARA
- B. Delta (parallel)
- C. Gantry
- D. Articulated

Q5. A robot returns to the same taught point within ± 0.02 mm every time, but that point is about 2 mm away from the commanded position. This robot has:

- A. Good accuracy and good repeatability
- B. Good accuracy but poor repeatability
- C. Poor accuracy but good repeatability
- D. Poor accuracy and poor repeatability

Q6. According to the lecture, which factor affects accuracy but does NOT affect repeatability?

- A. Temperature
- B. Payload
- C. Calibration
- D. Wear-out

Q7. Which of the following sensors gives a digital output signal?

- A. Resolver
- B. Potentiometer
- C. Optical encoder
- D. LVDT

Q8. For a DC motor, the speed is mainly controlled by ____, and the direction of rotation is determined by ____.

- A. the frequency; the duty cycle
- B. the supplied voltage; the polarity of the supplied voltage
- C. the duty cycle; the frequency

D. the motor current; the temperature

Q9. The pulse-width ratio (duty cycle) of a PWM signal is defined as:

A. t_{period}/t_{on}

B. t_{on}/t_{period}

C. V_{high}/V_{low}

D. $f \times t_{on}$

Q10. A robot must transfer a 15 kg pack of water bottles to a pallet. Which specification is the hard constraint that immediately rules out a small table-top robot?

A. Payload, with a margin for the gripper

B. Repeatability better than 0.001 mm

C. Tool speed faster than 10 m/s

D. More than 10 DOF

Section B: True/False (4 questions, 2 marks each, 8 marks)

Write T (True) or F (False) for each statement.

Q11. In PWM control of a DC motor, the supplied voltage is varied continuously to change the motor speed.

Q12. The H-bridge in a DC motor driver determines the direction of the motor current.

Q13. Wear-out of the transmission affects the repeatability of a robot but not its accuracy.

Q14. The sensitivity of a sensor is the ratio of the change in output to the change in input.

Section C: Calculation (4 questions, 10 marks each, 40 marks)

Q15. A robot is programmed to move to a tool position (100, 200, 0) several times. A laser scanner measures the tool position and obtains the following results (unit: mm):

$$(95, 203, 3), \quad (97, 205, 5), \quad (94, 207, 4), \quad (96, 206, 2), \quad (93, 204, 1)$$

Compute the accuracy and the repeatability of the robot.

Q16. A pressure sensor is calibrated by applying known pressures, with the following results:

Pressure P (kPa)	0	5	10	15	20
Output V (mV)	0	4	8	12	16

- (a) Determine the sensitivity of the sensor.
- (b) Estimate the output voltage when the applied pressure is 12.5 kPa.

Q17. A 24 V DC supply drives a motor through a PWM amplifier whose switching frequency is 20 kHz.

- (a) What is the average voltage applied to the motor when the duty cycle $D = 0.25$?
- (b) What duty cycle is needed to obtain an average voltage of about 18 V?
- (c) What is the switching period, and for how long is the switch ON within one period when $D = 0.25$?

Q18. A sensor is used to measure a constant input. Its sensitivity error is $\pm 0.1\%$ of the reading per $^{\circ}\text{C}$ change in ambient temperature. At an ambient temperature of 20°C the sensor outputs 4.00 V. Estimate the range of the output at an ambient temperature of 30°C , assuming the sensitivity error is the only source of error.

Section D: Short Answer (2 questions, 32 marks)

Q19. (12 marks) List at least six applications of industrial robots in manufacturing and logistics.

Q20. (20 marks) A pack of water bottles weighs 15 kg. The pack is transferred to a pallet, and the maximum distance between the pack and the pallet is 2 meters. What do you need to consider when selecting a robot for this task? Give details and reasons.

Part 2: Answer Key and Solutions 答案解析

Quick key (速查): 1-D, 2-C, 3-B, 4-C, 5-C, 6-C, 7-C, 8-B, 9-B, 10-A; 11-F, 12-T, 13-T, 14-T; 15: accuracy ≈ 7.98 mm, repeatability ≈ 2.40 mm; 16: (a) 0.8 mV/kPa, (b) 10 mV; 17: (a) 6 V, (b) 0.75, (c) 50 μs , 12.5 μs ; 18: 3.96–4.04 V.

Section A (选择题详解):

Q1. D. 焊接 (welding)、喷涂 (painting)、装配 (assembly)、拾取与放置 (pick and place)、产品检测 (product inspection)、测试 (testing)、抛光打磨 (polishing/grinding)、码垛 (palletizing)、上下料 (loading/unloading)、分拣 (sorting) 等都是讲义列出的典型应用; 餐厅做饭不是典型工业机器人应用。

Q2. C. 抛光自由曲面时, 工具需要 3 个自由度到达表面任意点 (定位), 再用 2 个自由度把工具轴线调成与表面垂直 (定向); 绕工具自身轴线的旋转 (roll) 不改变抛光接触, 所以最少 5 个自由度。实际中通常使用 6 自由度关节型机器人 (更通用, 可达任意姿态)。

Q3. B. SCARA 有 3 个旋转关节 (轴线互相平行且竖直)+ 1 个竖直移动的棱柱关节 = 4 DOF。讲义: SCARA 最少 4 轴, 竖直运动平滑快速, 水平面内高顺应性 (high compliance in the horizontal plane), 重复性 < 0.03 mm, 适合精密高速轻型装配。

Q4. C. 龙门机器人 (gantry) 工作空间非常大 (very large working envelope)、负载 10–1000 kg, 典型应用就是搬运超大件、长距离输送物料; 缺点是负载大时速度慢。关节型 (articulated) 负载可达 1–2000 kg 且速度快, 但工作空间是“相对体积大”而非绝对最大; SCARA 与 Delta 用于小型高速场合, 负载轻。

Q5. C. 每次回到同一示教点 $\pm 0.02 \text{ mm}$, 说明重复性 (repeatability) 很好; 但整体离指令位置差 2 mm , 说明精度 (accuracy) 差。精度相对指令位置 (commanded position), 重复性相对记录/示教位置 (recorded/taught position)。好重复性 + 差精度意味着系统性偏移 (systematic offset), 应检查标定 (calibration) 与机器人模型 (robot model)。

Q6. C. 影响精度的因素: 标定 (calibration)、机器人模型 (robot model)、运动控制 (motion control)、温度、负载、速度。影响重复性的因素: 温度、负载、速度、磨损 (wear-out)、运动控制。标定与机器人模型只影响精度; 磨损只影响重复性。

Q7. C. 光学编码器 (optical encoder) 输出数字量 (digital signal), 精度 < 1 角分; 旋转变压器 (resolver) 输出模拟量 (analog signal, 精度 > 2 角分); 电位计 (potentiometer) 与 LVDT 也都是模拟输出。

Q8. B. 讲义原话: 电机转速由供电电压 (supplied voltage) 决定, 转向由供电电压极性 (polarity) 决定。PWM 调节的是平均电压 (等效供电电压); H 桥决定电流方向; 电流环近似控制转矩 (torque), PID 再闭环修正速度/位置。

Q9. B. 占空比 (pulse-width ratio) $D = t_{on}/t_{period}$ 。PWM 不改变供电电压幅值, 而是以高频 (如 20 kHz) 通断, 理想平均电压约为 $D \times V_{supply}$ 。

Q10. A. 15 kg 的整包水加夹爪自重与安全裕量, 负载 (payload) 至少要 $15\text{--}20 \text{ kg}$, 直接排除小型桌面机器人; 其次看工作空间 (最大距离 2 m , 臂展必须覆盖) 与重复性 (码放到位)。B、C、D 的要求都不必要 (0.001 mm 、 10 m/s 、 10 DOF 均远超需求)。

Section B (判断题详解):

Q11. F. PWM 恰恰不连续改变供电电压: 电压幅值保持不变, 以一定脉冲比高速通断, 电机只“看到”平均效果, 理想平均电压 $V_{avg} \approx D \times V_{supply}$ 。

Q12. T. H 桥通过导通不同的开关组合改变流过电机的电流方向, 从而控制转向; PWM 调节平均电压控制转速; PID 对速度/位置环做闭环修正。

Q13. T. 磨损 (wear-out) 在重复性的影响因素里, 不在精度的影响因素里: 传动磨损使每次到达同一示教点的散布变大 (重复性变差), 但不改变系统整体的目标偏差 (精度)。

Q14. T. 灵敏度 (sensitivity) 是输出变化与输入变化之比 (output/input)。例: 热电偶灵敏度 $20 \text{ mV}/^\circ\text{C}$, 温度每变化 1°C 输出变化 20 mV 。

Section C (计算题详解):

【Q15 详解】

指令位置 (commanded position) $(X_c, Y_c, Z_c) = (100, 200, 0)$ 。RIA 公式: 先求每点相对指令点的误差距离 (偏差平方和再开方), 再取平均。

点 1 (95, 203, 3): 偏差 $(-5, 3, 3)$, $25 + 9 + 9 = 43$, $\sqrt{43} \approx 6.56 \text{ mm}$

点 2 (97, 205, 5): 偏差 $(-3, 5, 5)$, $9 + 25 + 25 = 59$, $\sqrt{59} \approx 7.68 \text{ mm}$

点 3 (94, 207, 4): 偏差 $(-6, 7, 4)$, $36 + 49 + 16 = 101$, $\sqrt{101} \approx 10.05 \text{ mm}$

点 4 (96, 206, 2): 偏差 $(-4, 6, 2)$, $16 + 36 + 4 = 56$, $\sqrt{56} \approx 7.48 \text{ mm}$

点 5 (93, 204, 1): 偏差 $(-7, 4, 1)$, $49 + 16 + 1 = 66$, $\sqrt{66} \approx 8.12 \text{ mm}$

$$\text{Accuracy} = \frac{6.56 + 7.68 + 10.05 + 7.48 + 8.12}{5} \approx \frac{39.90}{5} \approx 7.98 \text{ mm}$$

重复性: 均值 $\bar{X} = (95 + 97 + 94 + 96 + 93)/5 = 95$, $\bar{Y} = (203 + 205 + 207 + 206 + 204)/5 = 205$, $\bar{Z} = (3 + 5 + 4 + 2 + 1)/5 = 3$ 。每点相对均值的距离:

点 1 $(0, -2, 0): \sqrt{4} = 2.00 \text{ mm}$; 点 2 $(2, 0, 2): \sqrt{8} \approx 2.83 \text{ mm}$; 点 3 $(-1, 2, 1): \sqrt{6} \approx 2.45 \text{ mm}$; 点 4 $(1, 1, -1): \sqrt{3} \approx 1.73 \text{ mm}$; 点 5 $(-2, -1, -2): \sqrt{9} = 3.00 \text{ mm}$

$$\text{Repeatability} = \frac{2.00 + 2.83 + 2.45 + 1.73 + 3.00}{5} \approx \frac{12.01}{5} \approx 2.40 \text{ mm}$$

结论: 精度约 7.98 mm, 重复性约 2.40 mm。精度差于重复性, 说明机器人整体偏离目标较远 (可能有标定/模型误差), 但每次落点相对集中。

【Q16 详解】

(a) 灵敏度 (sensitivity)= 输出变化 / 输入变化。任取相邻两点: $\Delta V = 4 \text{ mV}, \Delta P = 5 \text{ kPa}$,

$$S = \frac{\Delta V}{\Delta P} = \frac{4 \text{ mV}}{5 \text{ kPa}} = 0.8 \text{ mV/kPa}$$

(b) 数据过原点且线性, $V = 0.8 \times P$ 。当 $P = 12.5 \text{ kPa}$ 时, $V = 0.8 \times 12.5 = 10 \text{ mV}$ 。

【Q17 详解】

(a) 理想平均电压约为占空比乘电源电压:

$$V_{avg} \approx D \times V_{supply} = 0.25 \times 24 = 6 \text{ V}$$

(b) 要求 $V_{avg} = 18 \text{ V}$, 则 $D = 18/24 = 0.75$ 。

(c) 开关周期 $T = 1/f = 1/20\,000 = 50 \mu\text{s}$; 导通时间 $t_{on} = D \times T = 0.25 \times 50 = 12.5 \mu\text{s}$ 。

【Q18 详解】

先澄清: 这里的 $^{\circ}\text{C}$ 是环境温度 (ambient temperature), 不是被测量——传感器测的是某个恒定输入 (讲义 p63-64 把温度列为环境因素 environmental requirements)。被测量没变, 所以名义输出仍是 4.00 V; 但环境温度从 20°C 升到 30°C 会让灵敏度漂移, 实际输出不确定, 只能给出范围。

讲义定义 (p58): 灵敏度误差是“占读数的百分比”—— $\pm 0.1\%$ of the reading per $^{\circ}\text{C}$, 误差基数就是读数本身 (不是满量程, 也不是灵敏度值)。

温度变化 $\Delta T = 30 - 20 = 10^{\circ}\text{C}$, 故总误差 = $\pm 0.1\% \times 10 = \pm 1\%$ of the reading。

题目中唯一的读数是 4.00 V, 所以把它当基数换算成电压误差:

$$4.00 \times (\pm 1\%) = \pm 0.04 \text{ V}$$

输出范围: $4.00 - 0.04 = 3.96 \text{ V}$ 到 $4.00 + 0.04 = 4.04 \text{ V}$, 即 3.96-4.04 V。

等价视角: 灵敏度随温度漂移 $\pm 1\%$, 同一输入下输出变为 $4.00 \times (1 \pm 0.01)$, 结果相同。讲义同款算法见 p53: 精度 $\pm 1\%$ of f.s.d., f.s.d. = 10 A $\rightarrow$ 误差 $\pm 0.1 \text{ A}$ ——百分比必须乘基准量才得到带单位的误差。

Section D (简答题详解):

【Q19 详解】

任写六个即满分: 焊接 (welding)、喷涂 (painting)、装配 (assembly)、拾取与放置 (pick and place)、产品检测 (product inspection)、测试 (testing)、抛光/打磨/去毛刺 (polishing/grinding/deburring)、机加工 (machining)、码垛 (palletizing)、上下料 (loading/unloading)、包装 (packaging)、分拣 (sorting)、仓储物流 (warehouse logistics, AGV/移动机器人搬运) 等。

【Q20 详解】

核心考虑 (给出理由):

1. 负载 (**Payload**): 水包 15 kg, 还要加上夹爪/吸盘自重与安全裕量, 机器人负载至少要 15-20 kg, 直接排除小型机器人;
2. 工作空间/臂展 (**Workspace/reach**): 最大搬运距离 2 m, 机器人臂展 (**reach**) 必须覆盖 2 m 及托盘区域, 否则够不到;
3. 自由度 (**DOF**): 码放需要定位加姿态调整, 至少 4 DOF(3 个定位 + 1 个竖直/旋转), 常用 6 自由度关节型;
4. 重复性 (**Repeatability**): 要准确码放到托盘上, 重复性通常要求 0.1 mm 量级或更好;
5. 速度/节拍 (**Speed/cycle time**): 决定单位时间码放数量, 影响生产效率;
6. 结构类型 (**Structure**): 6 轴关节型 (灵活、臂展大) 或龙门机 (大工作空间、重负载) 合适; SCARA/Delta 适合轻载高速场合;
7. 末端执行器 (**End-effector**): 夹爪或真空吸盘要适配整包水瓶, 其自重计入负载;
8. 环境与安全 (**Environment and safety**): 车间有水汽需防护等级 (IP rating); 人机共处需围栏或协作安全措施;
9. 其他: 成本、控制器与编程方式 (示教 vs 离线编程)、供电与占地。